



KEY 테스트

CRZ Technology





Mango-M32F2 KEY 테스트

KEY pinmap

■ KEY – PE11 / PE12

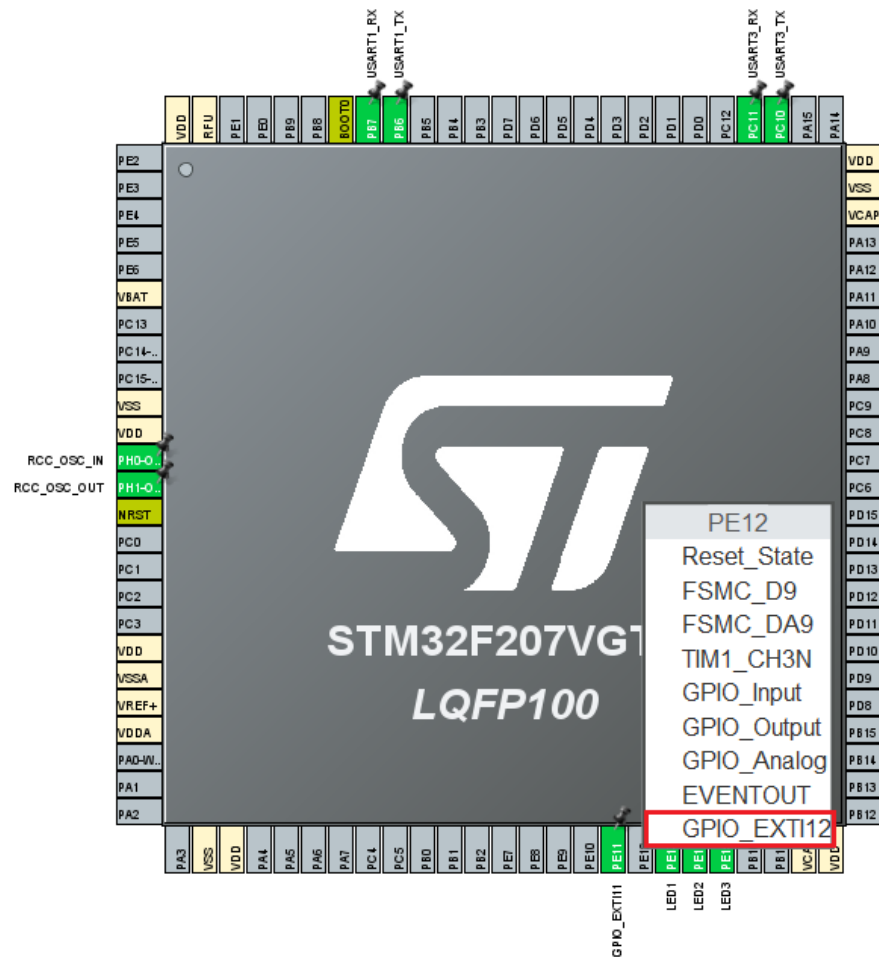
PE10	42	PE11
PE11	43	PE12
PE12	44	PF13

PE11	KEY1
PE12	KEY2
PF13	IFD1

■ PE11 / PE12 – GPIO_EXTI12 선택.

 Pinout view

 System view



GPIO 설정

- “System Core“ 클릭, “GPIO” 클릭.
- KEY를 누르면 LOW가 되므로 Falling edge trigger로 설정.

Device Configuration Tool
key_prj.toc - Pinout & Configuration

Pinout & Configuration | Clock Configuration | Project Manager

Software Packs | Pinout

GPIO Mode and Configuration

Configuration

Group By Peripherals

GPIO | RCC | USART | NVIC

Search Signals
Search (Ctrl+F)

Pin | Signal | GPIO output level | GPIO mode | GPIO Pull-up/Pull-down | Maximum output | User Label

PE11	n/a	n/a	External Interrupt Mode with Falling edge trigger detect	No pull-up and no pull-down	Low	KEY1
PE12	n/a	n/a	External Interrupt Mode with Falling edge trigger detect	No pull-up and no pull-down	Low	KEY2
PE13	n/a	High	Output Push Pull	No pull-up and no pull-down	Low	LED1
PE14	n/a	High	Output Push Pull	No pull-up and no pull-down	Low	LED2
PE15	n/a	High	Output Push Pull	No pull-up and no pull-down	Low	LED3

PE11 Configuration :

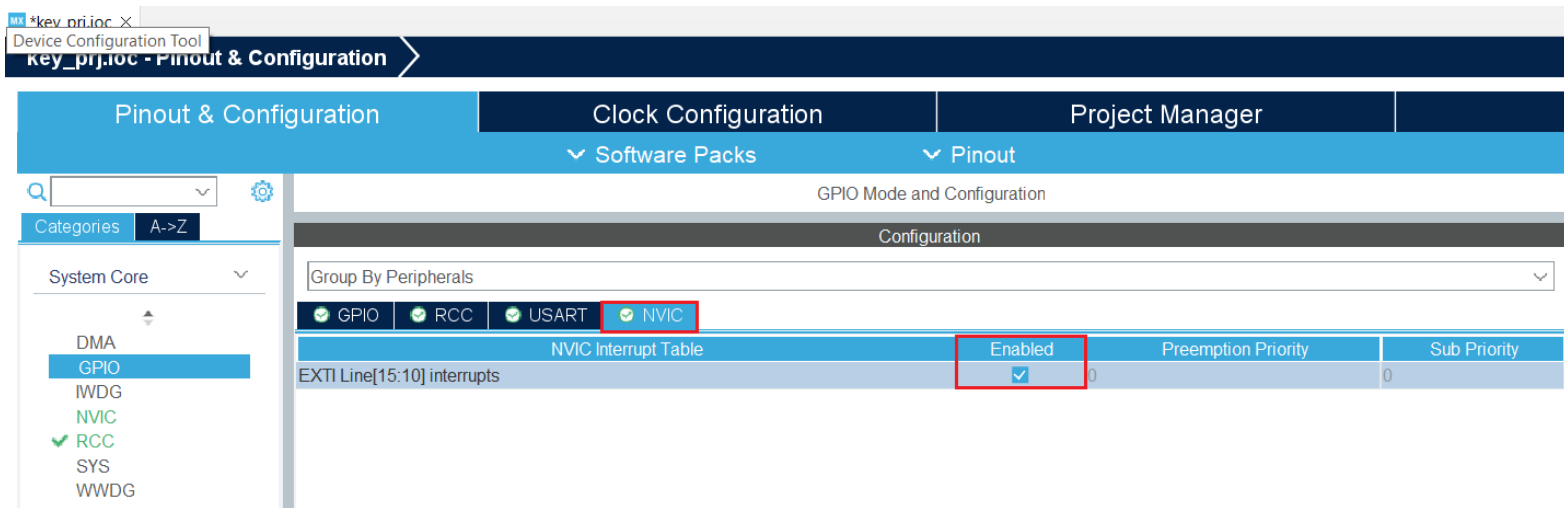
GPIO mode: External Interrupt Mode with Falling edge trigger detection

GPIO Pull-up/Pull-down: No pull-up and no pull-down

User Label: KEY1

GPIO 설정

- “System Core“ 클릭, “GPIO” 클릭.
- NVIC tab에서 EXTI Line[15:10] interrupts Enable.



KEY Test Project

- KEY1 Press시 LED1 토글, KEY2 Press시 LED2 토글되도록 구현.
- core/inc/main.h

```
59  /* Private defines -----*/
60  #define KEY1_Pin GPIO_PIN_11
61  #define KEY1_GPIO_Port GPIOE
62  #define KEY1_EXTI_IRQn EXTI15_10_IRQn
63  #define KEY2_Pin GPIO_PIN_12
64  #define KEY2_GPIO_Port GPIOE
65  #define KEY2_EXTI_IRQn EXTI15_10_IRQn
66  #define LED1_Pin GPIO_PIN_13
67  #define LED1_GPIO_Port GPIOE
68  #define LED2_Pin GPIO_PIN_14
69  #define LED2_GPIO_Port GPIOE
70  #define LED3_Pin GPIO_PIN_15
71  #define LED3_GPIO_Port GPIOE
72  ..
```

KEY Test Project

■ core/src/stm32f2xx_it.c

```
201- /**
202-  * @brief This function handles EXTI Line[15:10] interrupts.
203-  */
204- void EXTI15_10_IRQHandler(void)
205- {
206-     /* USER CODE BEGIN EXTI15_10_IRQn 0 */
207-
208-     /* USER CODE END EXTI15_10_IRQn 0 */
209-     HAL_GPIO_EXTI_IRQHandler(KEY1_Pin);
210-     HAL_GPIO_EXTI_IRQHandler(KEY2_Pin);
211-     /* USER CODE BEGIN EXTI15_10_IRQn 1 */
212-
213-     /* USER CODE END EXTI15_10_IRQn 1 */
214- }
215
```


KEY Test Project

■ Drivers/STM32F2xx_HAL_Driver/src/stm32f2xx_hal_gpio.c

```
489- /**
490  * @brief This function handles EXTI interrupt request.
491  * @param GPIO_Pin Specifies the pins connected EXTI line
492  * @retval None
493  */
494- void HAL_GPIO_EXTI_IRQHandler(uint16_t GPIO_Pin)
495  {
496      /* EXTI line interrupt detected */
497      if(__HAL_GPIO_EXTI_GET_IT(GPIO_Pin) != 0x00u)
498      {
499          __HAL_GPIO_EXTI_CLEAR_IT(GPIO_Pin);
500          HAL_GPIO_EXTI_Callback(GPIO_Pin);
501      }
502  }
503
504- /**
505  * @brief EXTI line detection callbacks.
506  * @param GPIO_Pin Specifies the pins connected EXTI line
507  * @retval None
508  */
509- __weak void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
510  {
511      /* Prevent unused argument(s) compilation warning */
512      UNUSED(GPIO_Pin);
513-      /* NOTE: This function Should not be modified, when the callback is needed,
514       the HAL_GPIO_EXTI_Callback could be implemented in the user file
515       */
516  }
517
```

KEY Test Project

■ core/src/main.c, HAL_GPIO_EXTI_Callback() 재정의

```
71
72 void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
73 {
74     if(GPIO_Pin == KEY1_Pin)
75     {
76         if(HAL_GPIO_ReadPin(KEY1_GPIO_Port, KEY1_Pin) == GPIO_PIN_RESET)
77         {
78             HAL_GPIO_TogglePin(LED1_GPIO_Port, LED1_Pin );
79         }
80     }
81     else if(GPIO_Pin == KEY2_Pin)
82     {
83         if(HAL_GPIO_ReadPin(KEY2_GPIO_Port, KEY2_Pin) == GPIO_PIN_RESET)
84         {
85             HAL_GPIO_TogglePin(LED2_GPIO_Port, LED2_Pin );
86         }
87     }
88 }
89 /* USER CODE END 0 */
```

Test 결과 확인

- KEY1 Press시 LED1 토글 확인.
- KEY2 Press시 LED2 토글 확인.

Thank You

